KU3VACMAT202: MATHEMATICAL LOGIC

MODULE IV : Logic and Propositional Calculus III

10.10 Logical Implication

What does "logical implication" really mean? In everyday language, when we say "if it rains, the ground gets wet", we mean the second fact follows from the first. In logic, we make this precise: P logically implies Q when there is no possible situation (no row of the truth table) in which P is true and Q is false. Logical implication is about the *form* of the statements, not their subject matter.

A proposition $P(p, q, \ldots)$ is said to **logically imply** a proposition $Q(p, q, \ldots)$, written

$$P(p, q, \ldots) \Rightarrow Q(p, q, \ldots)$$

if $Q(p, q, \ldots)$ is true whenever $P(p, q, \ldots)$ is true.

Example 10.8. We claim that p logically implies $p \vee q$. For consider the truth table in Fig. 10-11. Observe that p is true in Cases (rows) 1 and 2, and in these cases $p \vee q$ is also true. Thus $p \Rightarrow p \vee q$.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Fig. 10-11

Real-life illustration. Let p be "It is raining" and q be "It is cloudy". Then $p \vee q$ is "It is raining or it is cloudy". If it is indeed raining, then "It is raining or it is cloudy" is automatically true — it does not matter whether it is cloudy. So "It is raining" logically implies "It is raining or it is cloudy". This is the pattern behind the logical rule called *addition* (or *or-introduction*): from any statement P, infer $P \vee Q$.

Now if $Q(p, q, \ldots)$ is true whenever $P(p, q, \ldots)$ is true, then the argument

$$P(p, q, \ldots) \vdash Q(p, q, \ldots)$$

is valid; and conversely. Furthermore, the argument $P \vdash Q$ is valid if and only if the conditional statement $P \to Q$ is always true, i.e., a tautology. We state this result formally.

Theorem 10.4. For any propositions $P(p, q, \ldots)$ and $Q(p, q, \ldots)$ the following three statements are equivalent:

(i) $P(p, q, \ldots)$ logically implies $Q(p, q, \ldots)$.

(ii) The argument $P(p, q, \ldots) \vdash Q(p, q, \ldots)$ is valid.

(iii) The proposition $P(p, q, \ldots) \to Q(p, q, \ldots)$ is a tautology.

Why the theorem matters. The theorem gives three equivalent ways to check the same thing — semantic (truth tables), syntactic (valid arguments), and truth-table (tautology). For instance, to check whether "If a number is divisible by 4, then it is even" is a valid implication, we can reason directly, test the argument form, or build a truth table. All three routes lead to the same verdict.

We note that some logicians and many texts use the word "implies" in the same sense as we use "logically implies", and so they distinguish between "implies" and "if . . . then". These two distinct concepts are, of course, intimately related as seen in the above theorem.

Gafoor I — Assistant Professor

Department of Mathematics